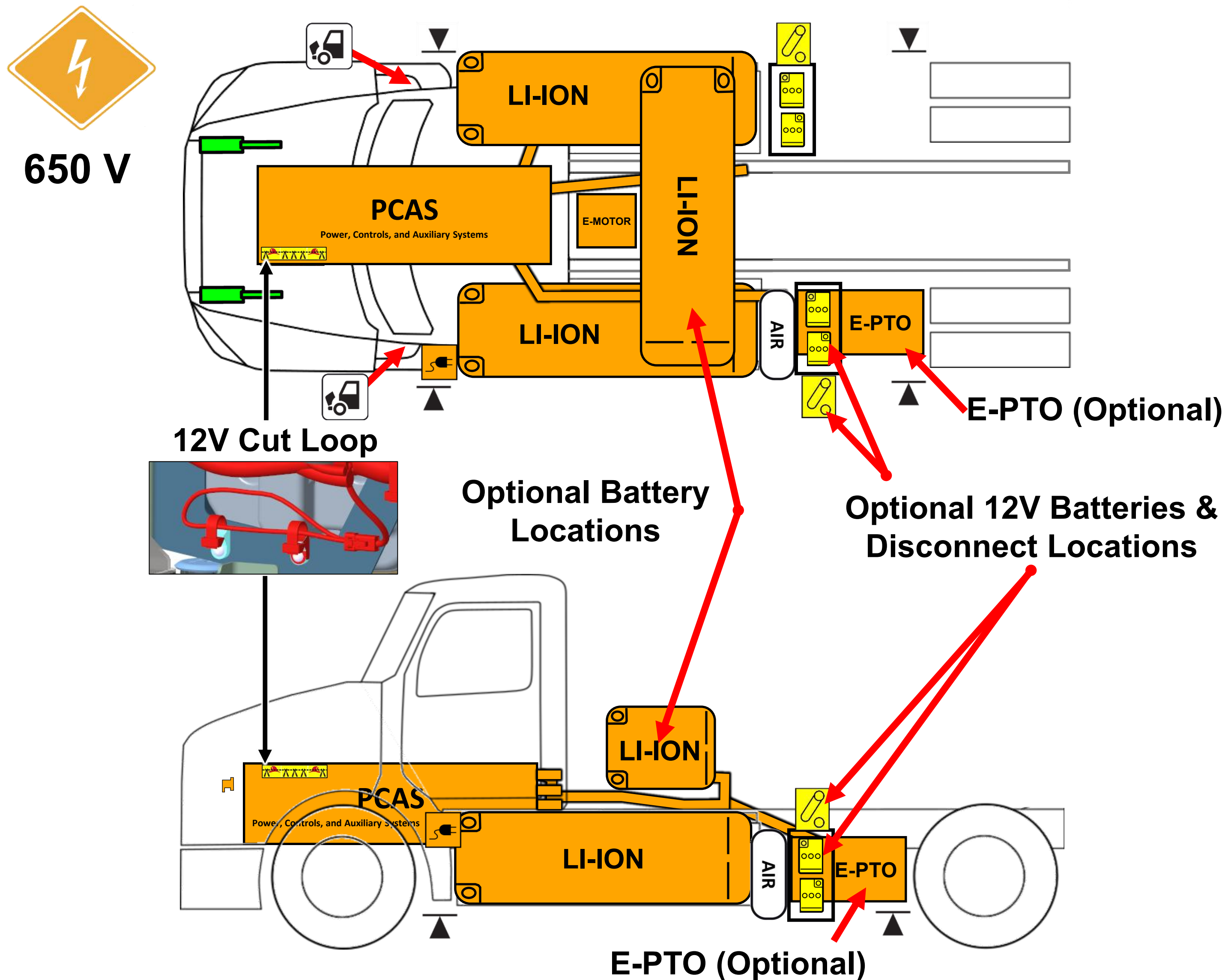




Model Year: 2025 – Current
Release Date: 12/08/2025
Part Number: Y53-6248-1B1



Warning: Check for labels identifying any additional High Voltage components added by body builders.



High Voltage Li-Ion Battery Pack



High Voltage Cables



High
Voltage
Region



Relay Box
Includes MSD (Manual
Service Disconnect)



Emergency Cable Cut Loop

Charger
Inlet

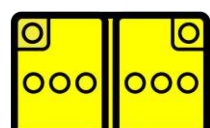
Lift Point



Hood
Release



Gas
Strut



Low Voltage Batteries



12V
Disconnect



Compressed Air Tank



Height
Control

Part Number: Y53-6248-1B1
Release Date: 12/08/2025

Page 1 of 7

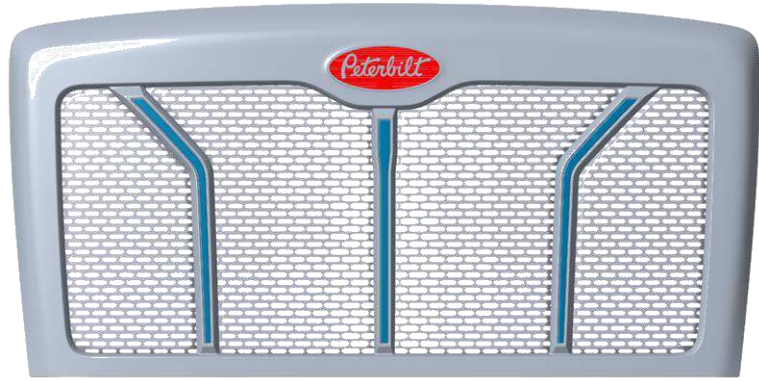
1. Identification / Recognition



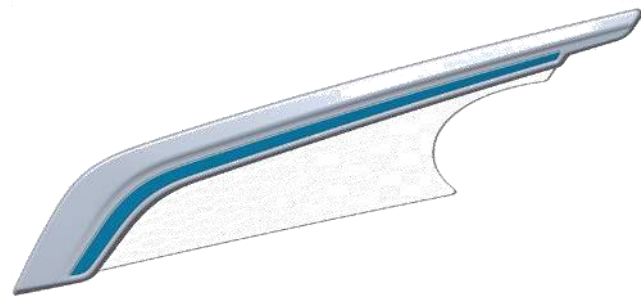
Warning: Always wear full fire fighter PPE (turnout gear), including a positive pressure self-contained breathing apparatus, when approaching this vehicle.

Peterbilt Exterior Identification

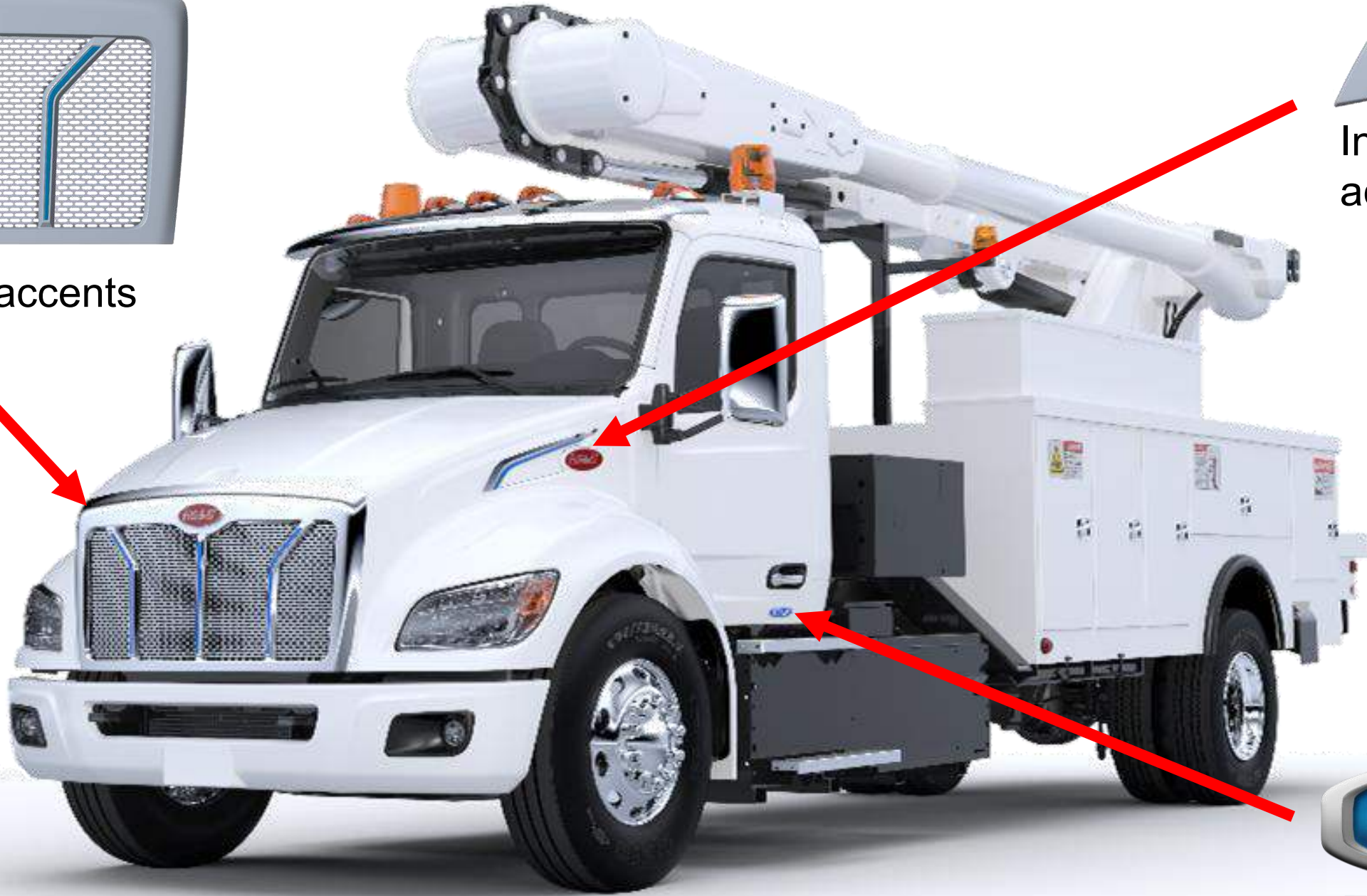
536EV, 537EV, and 548EV



Grille with Blue accents



Intake Shield with Blue accent



EV badge on both doors

Kenworth Exterior Identification

T280E, T380E, T480E



Intake Shield



Grille with Blue Kenworth badge



EV badge under both exterior mirrors

2. Immobilization / Stabilization / Lifting

- 

Warning: Keep all rescue equipment clear of high voltage components with a recommended clearance of 12 inches (30 cm).
- 

Warning: Vehicle noise may be reduced in some operation modes. Failure to shutdown the truck before immobilization could result in death, severe injury, or property damage.
- 

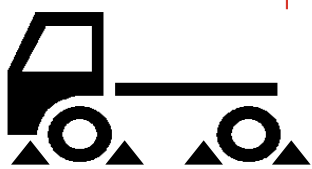
Warning: Assume all high voltage components are always energized. Do not cut any High Voltage components, including orange high voltage cables.

Immobilization



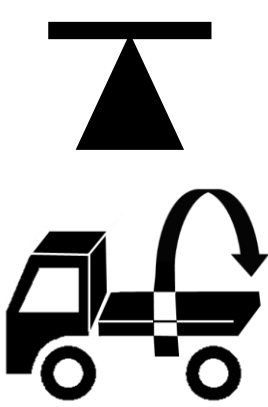
- Step 1:** Unplug the charger or remove power from the charger.
- Step 2:** Ensure the vehicle is in Neutral by turning the gear selector on the right stalk to N.
- Step 3:** Remove the key from the ignition.
- Step 4:** Engage tractor/truck park brake by pulling out switch.
- Step 4b (if applicable):** Engage trailer air brake by pulling out switch.

Stabilization



Block all the wheels using wheel chocks

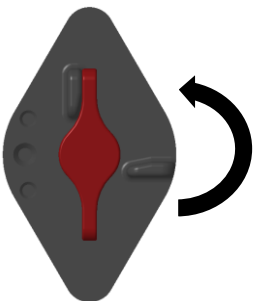
Lifting



- It is highly recommended to use wheel lifts or platform lifts, but axle lifts may be used as well. Do NOT lift from any HV components.
- To rotate the truck, wrap chains around both axles to rotate the truck to an upright stable position.

3. Disable Direct Hazards / Safety Regulations

Disabling High Voltage



Step 1: Go to low voltage battery box on the side of the truck.

Step 2:
(Primary Step): Turn 12V Disconnect Counterclockwise to OFF Position.



(Alternate Step): Cut a 5-inch (13 cm) segment (2 cuts) from the red cut loop (identified in the Rescue Sheet Diagram).

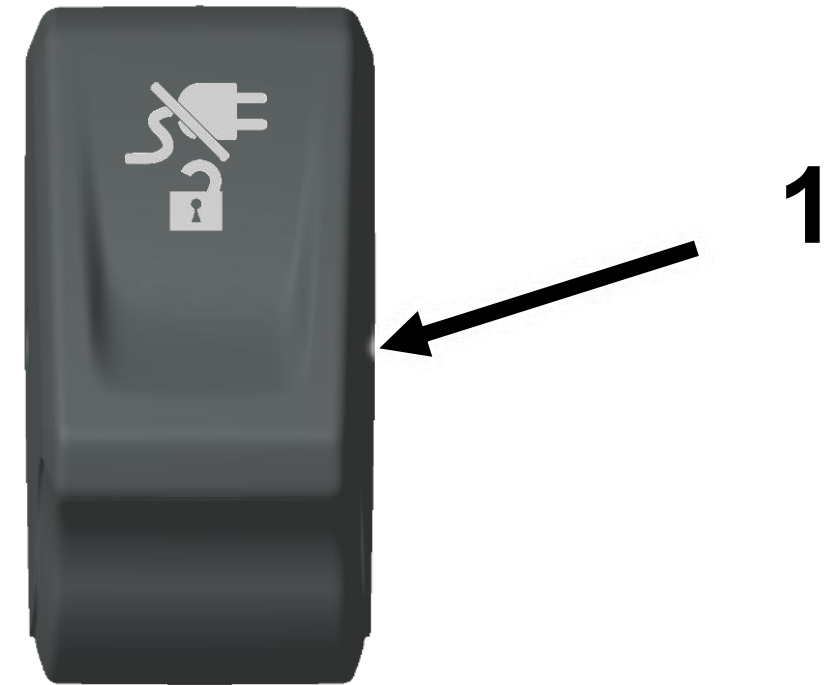


Step 3: Wait 2 Minutes for High Voltage Capacitors to Discharge.

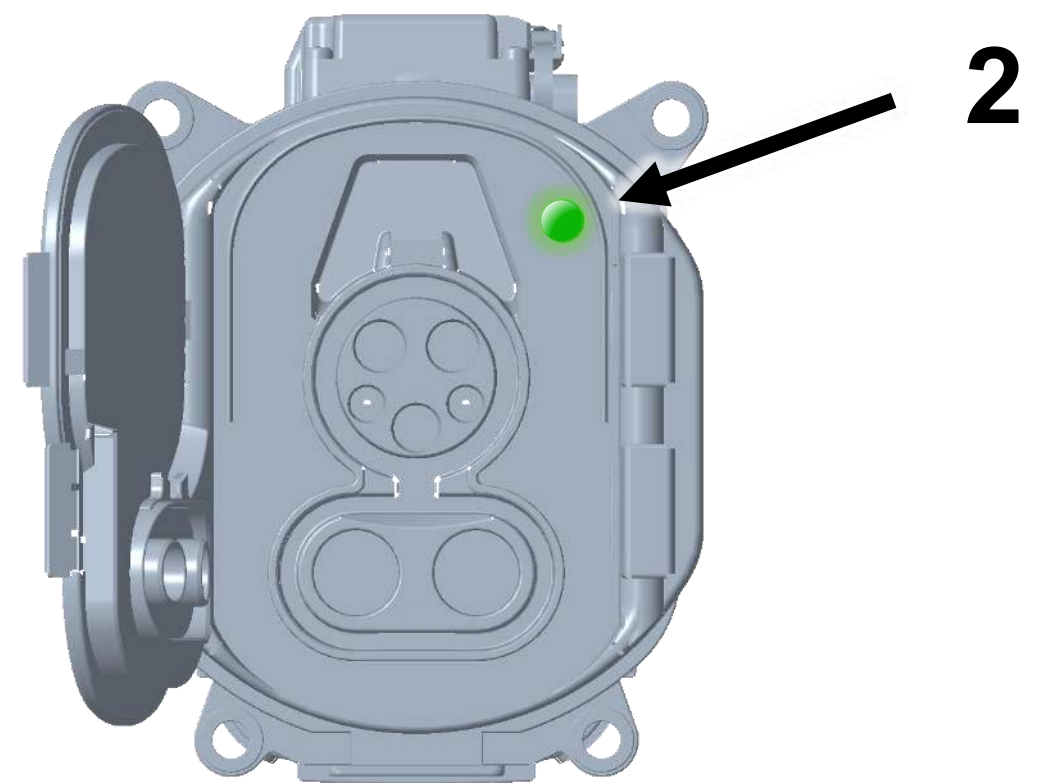
3. Disable Direct Hazards / Safety Regulations (continued)

If Truck is Charging

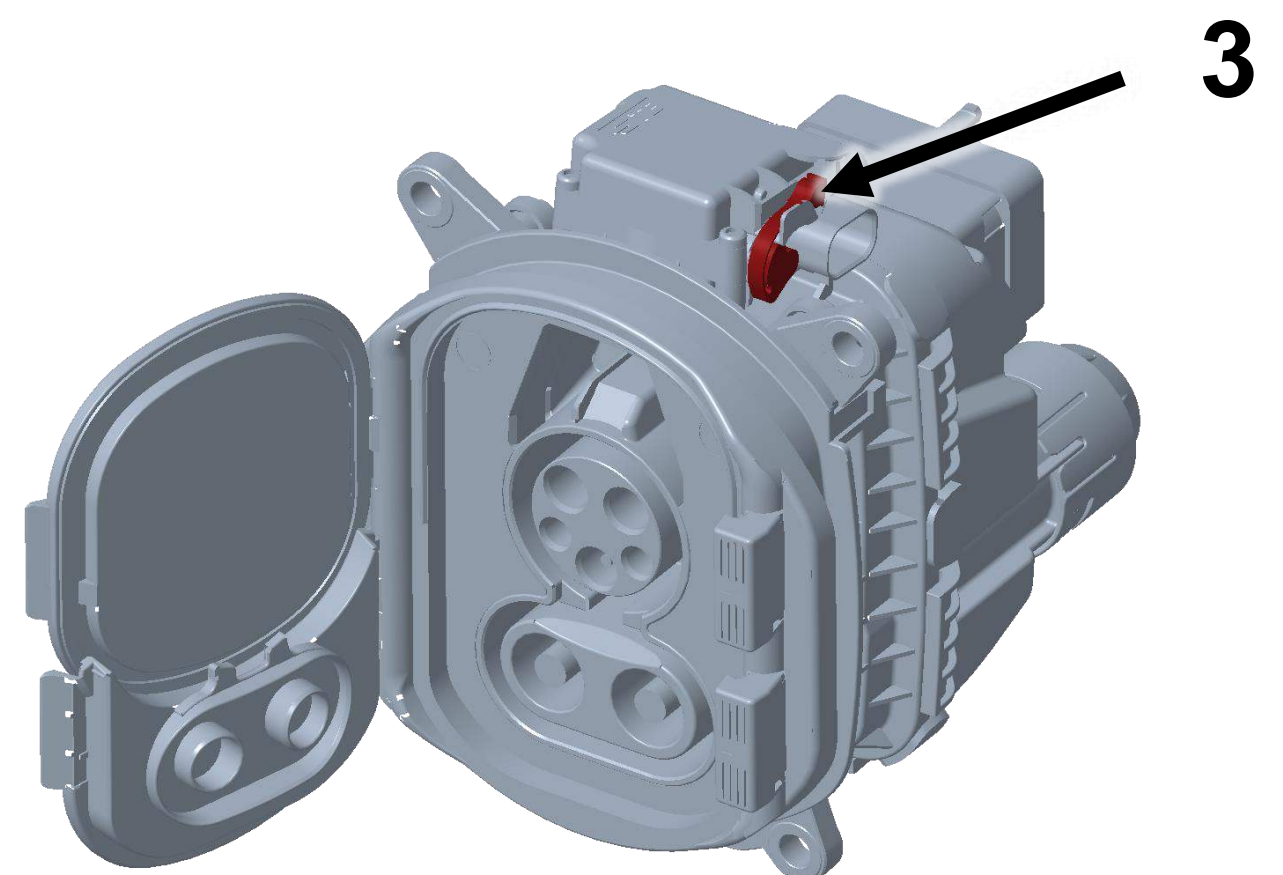
Step 1: Locate Stop Charging Switch (1) on the dashboard and toggle switch to begin the stop charging function.



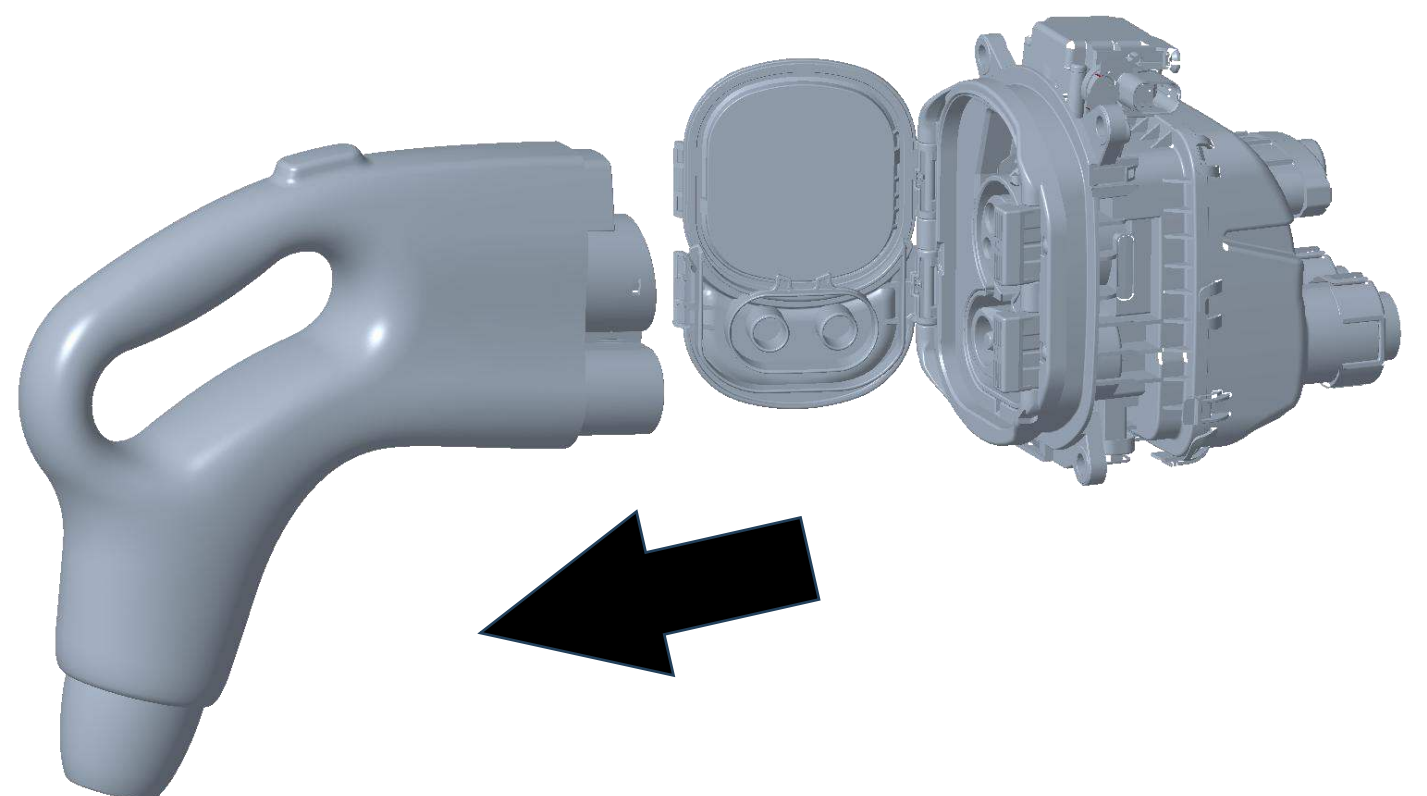
Step 2: Wait for charge indication light (2) on the charge inlet to show a solid green light indicating that the charge plug is unlocked.



Alternate Step: If charge stop function fails or charge inlet does not unlock. Locate emergency unlock lever (3) on charge inlet and pull up.



Step 3: Remove charge plug from charge inlet port with a pulling force.

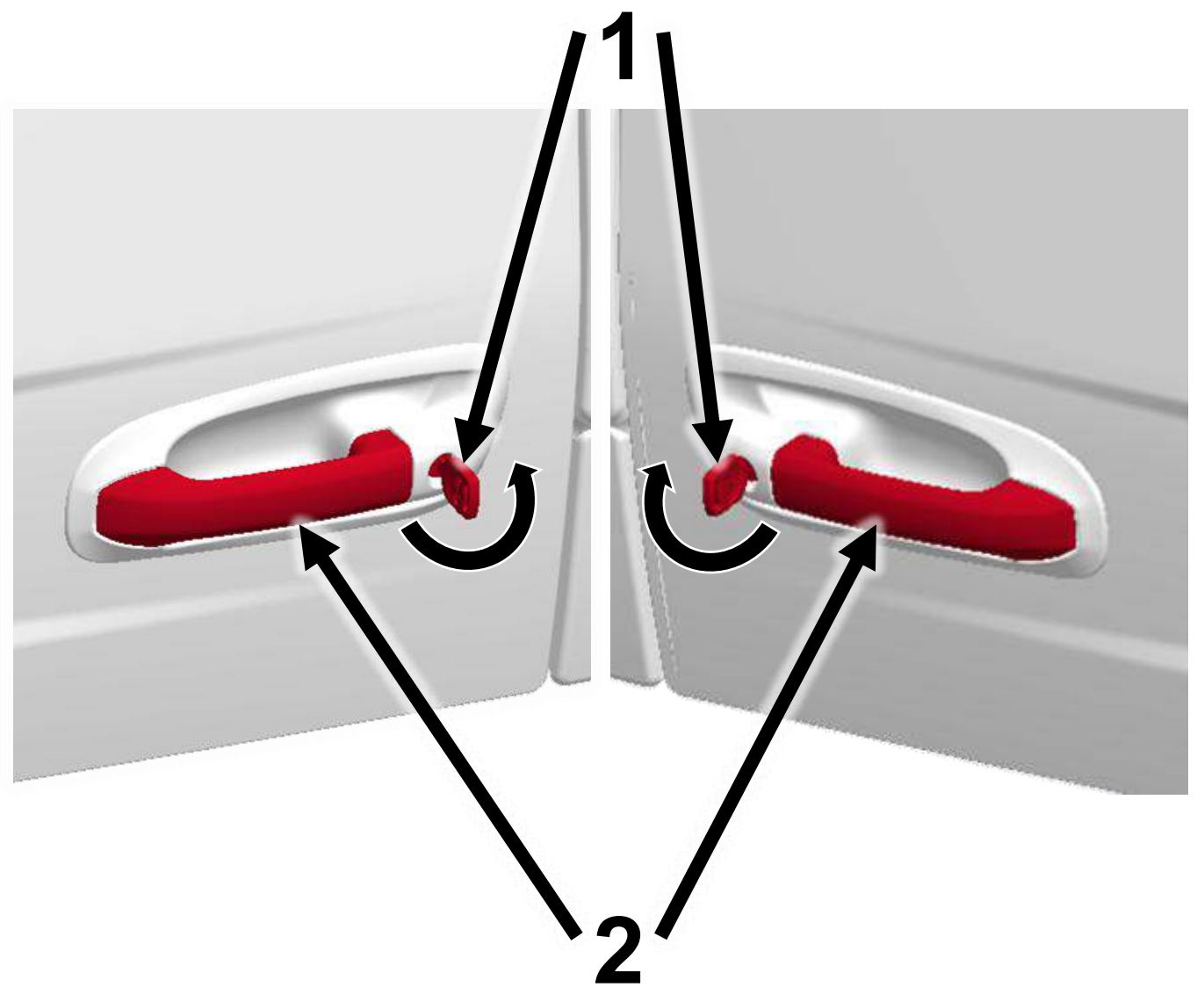


4. Access to the Occupants

Opening Doors from the Outside

Step 1: Insert key (1) into door lock turning key anticlockwise for left-hand door or clockwise for right-hand door to unlock

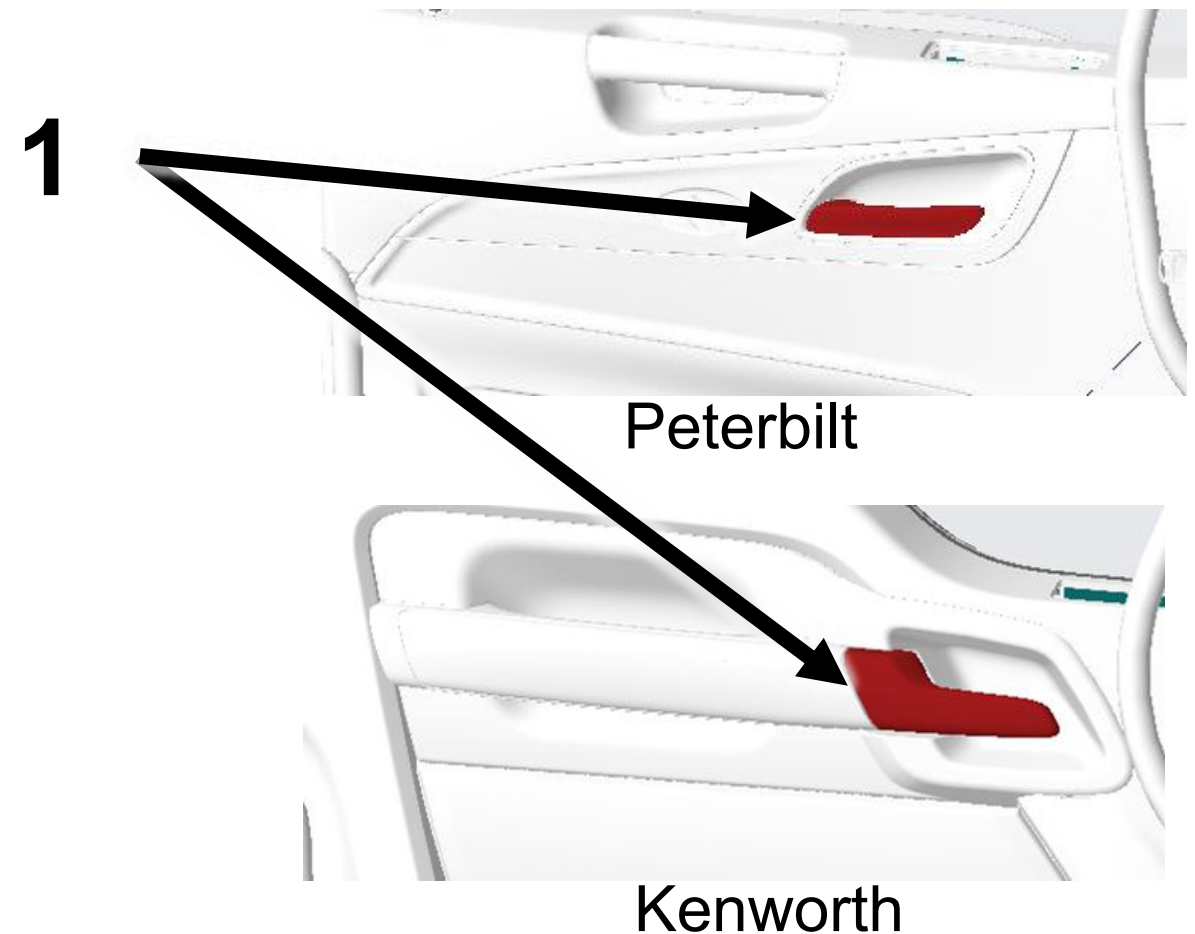
Step 2: Grasp door handle (2) and pull out while exerting some force on the door in the outward direction



Opening Doors from the Inside

Step 1: Pull on door handle (1) to release door latch

Step 2: Firmly push door outward



Seat Adjustment

Up/Down Adjustment: Pull switch (1) up or down to adjust seat up or down

Forward/Backward Adjustment: Pull handle 2 upward and push seat to slide forward or backward



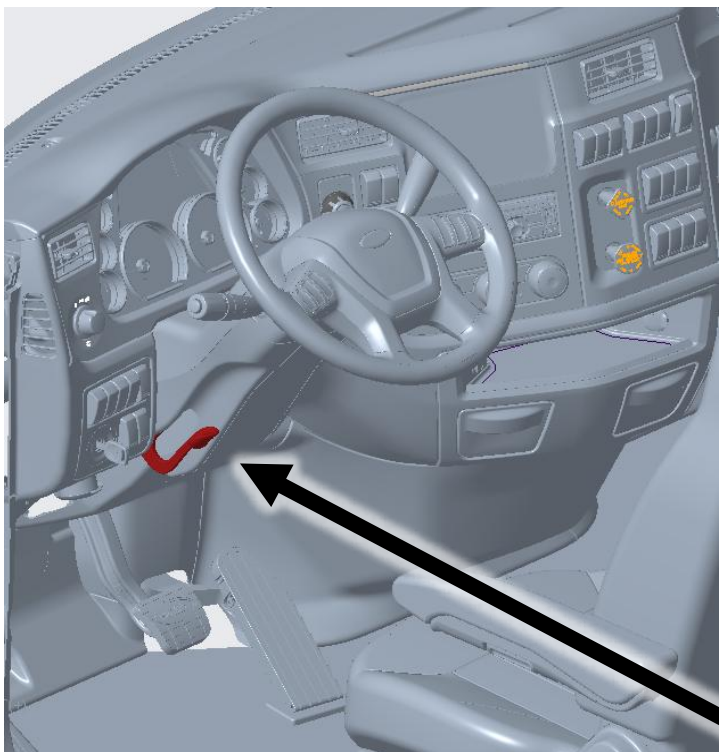
4. Access to the Occupants (continued)

Steering Column Adjustment (if applicable)

Step 1: Pull lever (1) downward to unlock steering adjustment

Step 2: To adjust steering column up/down, pull steering wheel up or down, to move steering wheel forward or backward, push or pull steering wheel

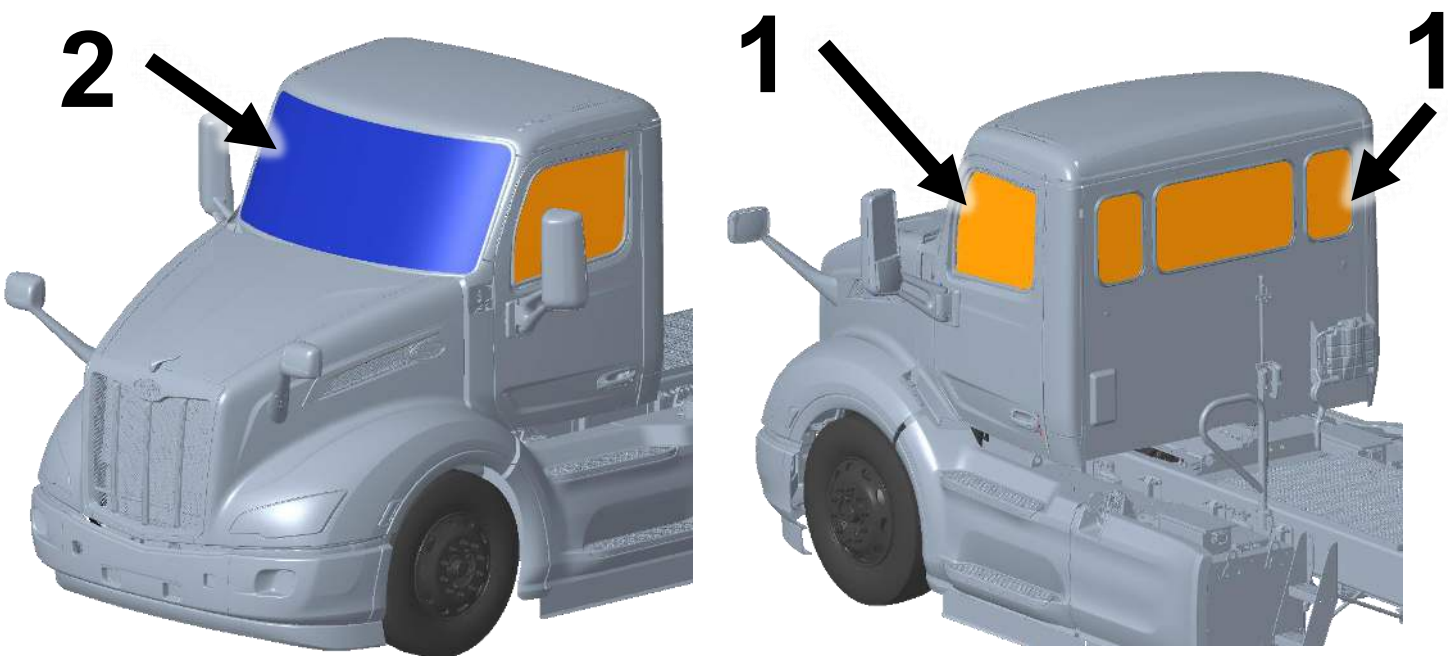
Step 3: Push lever (1) upward to lock steering wheel in new location



Window and Windshield Material

All glass materials listed below by location:

- Side Windows:** Tempered glass (1)
- Rear Windows:** Tempered glass (1)
- Windshield:** Laminated glass (2)



Cab Material

Note: There is no high strength or ultra-high strength steel present in the cab. The cab is predominantly made of aluminum material.



5. Stored Energy / Liquids / Gases / Solids



High Voltage (650 V)



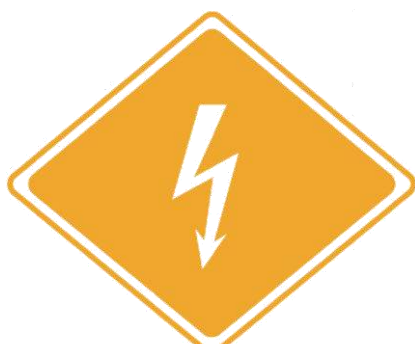
Corrosives



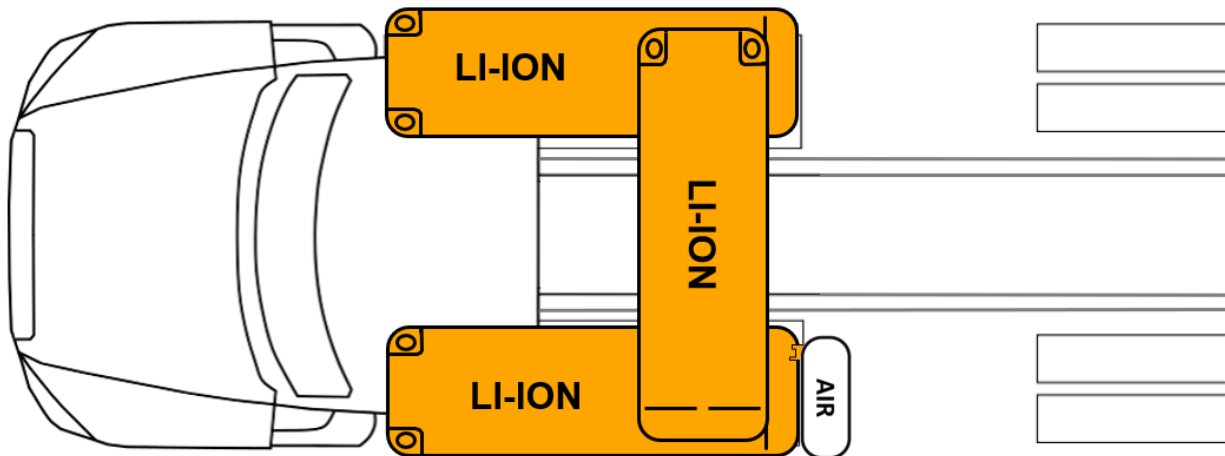
Flammable



Health Hazards



650 V



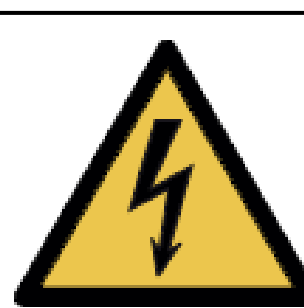
Air tanks and system hoses may have an air pressure of 100-130 psi (689-897 kPa).



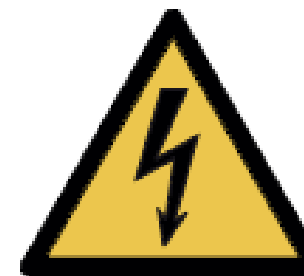
High voltage components are cooled with a glycol-based automotive coolant. This liquid is red and may leak if components are damaged. The vehicle also carries R134a refrigerant

Please contact local and state authorities for more information regarding proper response and clean up of hazardous materials.

6. In Case of Fire

 Warning: Always wear full fire fighter PPE (turnout gear), including a positive pressure self-contained breathing apparatus.  Warning: Treat fires involving charging stations as energized fires until power to the charger can be shut down.  Warning: There is always risk of reignition	
Use Large Amounts of Water to Extinguish Li-ion Fires	Do Not Use Wet Foam
Hazardous to Human Health: • May cause an allergic skin reaction. • Do not breathe dust, fumes, gas, mist, vapors, or spray.	Flammable Components
Explosion Hazard: • Explosive gas could accumulate. • Move truck outside building after extinguishing fire.	Corrosives: • Causes skin burns and eye damage.
High Voltage (650 V): • CAT III (1000 V) rated gloves required for exposed HV parts.	Check Li-ion Battery Pack for Unexpected Rising Temps with Thermal Imaging Camera (TIC or IR Gun)

7. Water Submersion

	If the vehicle has been submerged there is a chance of secondary impact damage to high voltage components around the vehicle. These components present a hazard and should only be handled while wearing the correct PPE. If the vehicle does NOT have impact damage, there is a low electrical shock risk. Remove the vehicle from the water. Let the water drain and follow the immobilization steps in Section 3. Do NOT attempt to drive.
--	---

8. Towing / Transportation / Storage

Towing/Transport	
Towing Method	Towing: Towing must be done with the rear wheels off the ground. In a scenario where this is not possible, the driveshaft between the drive motor and rear axle must be removed and the motor must be in neutral. If the driveshaft is not removed the motor can generate electricity. This presents a potential hazard. If any of the high voltage components were damaged, transport the truck with all the wheels off the ground. Do NOT attempt to drive. Transportation: If transporting the vehicle with a damaged high voltage system, periodically check the batteries and other high voltage components to make sure the vehicle stays cold throughout the duration of the trip. Storage:  Damaged vehicles must be inspected to ensure there is no chance of reignition. Store outdoors 50 feet (15 m) away from other equipment/structures and routinely check the battery pack for rising temperatures with a Thermal Imaging Camera Emergency: If first responders are unable to safely remove the driveshaft for towing, the truck can be moved at a max speed of 1.5 mph (2.4 kph) for a max distance of 0.5 mi (0.8 km).